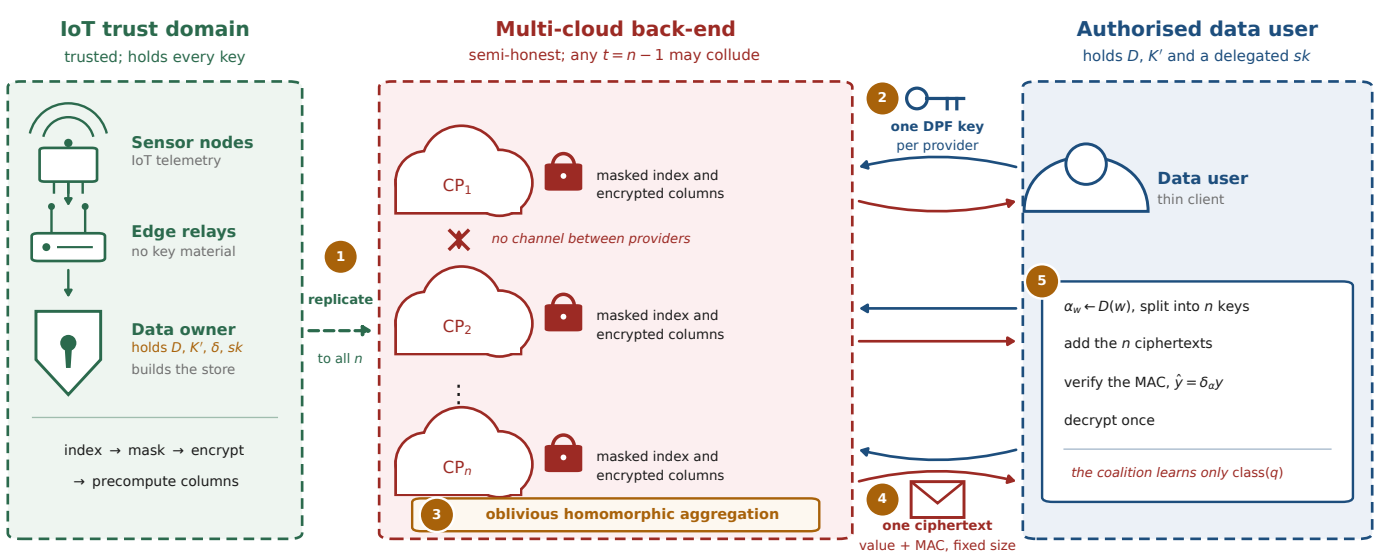
IoT trust domain
trusted; holds every key
Sensor nodes
IoT telemetry
Edge relays
no key material
Data owner
holds D, K′, δ, sk
builds the store
index → mask → encrypt
→ precompute columns
1
replicate
to all n
Multi-cloud back-end
semi-honest; any t = n − 1 may collude
CP1
masked index and encrypted columns
no channel between providers
CP2
masked index and encrypted columns
CPn
masked index and encrypted columns
3
oblivious homomorphic aggregation
2
one DPF key
per provider
4
one ciphertext
value + MAC, fixed size
Authorised data user
holds D, K′ and a delegated sk
Data user
thin client
5
αw ← D(w), split into n keys
add the n ciphertexts
verify the MAC, ŷ = δα y
decrypt once
the coalition learns only class(q)